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"Comparative Evaluation of Wearable Devices for Stress Prediction: Empatica E4 vs. Polar H10"

1.0 Executive Summary

This study addresses the growing importance of wearable health monitoring devices that aim to match the reliability and accuracy of clinical-grade systems. It presents a comparative evaluation of two widely used wearable devices, the Empatica E4 wristband and the Polar H10 chest strap, with a specific focus on stress monitoring. Biosignal data from both devices were systematically collected, processed, and analyzed to assess their effectiveness in real-world health monitoring scenarios. Stress levels were categorized into low, medium, and high classes based on predefined threshold values derived from the collected physiological signals. Machine learning techniques were employed to evaluate device performance, with particular emphasis on classification accuracy. The CatBoost Classifier was used as the primary predictive model for stress level detection. Experimental results indicate that the Polar H10 achieved higher predictive accuracy compared to the Empatica E4. These findings highlight the superior reliability of chest-strap–based sensing for stress prediction. Overall, the study provides valuable insights into wearable device performance and contributes to future research and development in wearable health technology and stress monitoring systems.

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2.0 Introduction, Problem Statement and Objectives

2.1 Introduction

The rapid advancement of wearable technology has significantly transformed the healthcare domain by enabling continuous and non-invasive monitoring of physiological parameters. Wearable health devices are increasingly used for tracking vital signals related to physical activity, emotional well-being, and stress levels in everyday life. Unlike traditional clinical equipment, these devices are designed to be portable, user-friendly, and suitable for long-term monitoring outside hospital environments. Stress monitoring has gained particular importance due to its strong association with various physical and mental health conditions. Accurate detection of stress through wearable biosignals can support early diagnosis and timely intervention. However, the reliability and accuracy of different wearable devices vary based on sensor type, placement, and signal quality. This study focuses on evaluating and comparing two popular wearable devices, the Empatica E4 wristband and the Polar H10 chest strap. By analyzing their biosignal data using machine learning techniques, the work aims to identify the device that provides more reliable stress prediction. The outcomes of this research are expected to support informed device selection and guide future developments in wearable health monitoring system.

2.2 Problem Statement

Wearable devices are widely used to monitor stress by collecting physiological signals, but their accuracy and reliability differ from one device to another. There is a lack of clear understanding about which type of wearable device provides more accurate stress prediction in real-world conditions. Wrist-based and chest-based devices capture biosignals in different ways, which may affect the quality of stress monitoring. Without proper comparison, it is difficult to select the most suitable device for reliable stress detection. Therefore, the problem is to evaluate and compare the performance of the Empatica E4 wristband and the Polar H10 chest strap for stress monitoring using machine learning techniques, and to identify which device produces more accurate and dependable stress level predictions

2.3 Objectives

1. To develop a framework that contrasts physiological signals acquired by Empatica E4 and Polar H10 for efficient detection of stress.
2. To develop a pre-processing pipeline that resamples, merges, handles missing data, and extracts features to prepare quality input data for analysis.
3. To implement supervised machine learning models, such as Random Forest, Logistic Regression, Decision Tree, XGBoost, CatBoost Classifiers, LightGBM Classifier, MLP Classifier, and Deep MLP Classifier, for accuracy in stress classification levels and device-to-device comparison.
4. To evaluate supervised models using metrics such as Accuracy and Weighted F1-score for effective stress detection between Empatica E4 and Polar H10.

3.0 Literature Review and Research Gap Analysis

The studies that have been reviewed are divided into two primary research categories. The former group encompasses the works that put heavy emphasis on cardiovascular monitoring via the use of wearable devices, and particular emphasis on HR and HRV as the key physiological indicators. The second one consists of the literature covering the aspect of sleep monitoring, where wearable sensors are tested in the context of their potential to identify the sleep pattern and the other way of categorizing sleep stages reliably. These two paths taken together constitute the two major themes of the wearable validation literature and give a logical structure to how previous work can be examined.

Most studies focus on either cardiovascular or sleep monitoring separately, instead of integrating both domains. Existing research evaluates single devices or single signals, with limited comparison across multiple wearable devices. The impact of sensor placement and device design on signal reliability is not fully studied. There is a lack of studies comparing multimodal wearable devices (e.g., Empatica E4 vs Polar H10) for stress prediction.

4.0 Methodology

1. Wearable technology is increasingly used for continuous health monitoring and tracking physiological parameters.
2. Different wearable devices vary in architecture, sensor design, and signal-processing algorithms, making device selection difficult for specific health applications.
3. ECG-based chest-strap devices are preferred for accurate heart rate variability (HRV) measurement due to their high signal quality.
4. Wrist-worn devices are commonly used for stress studies as they capture signals such as blood volume pulse (BVP) and skin temperature.
5. This study conducts a systematic analysis of two widely used wearables: Empatica E4 (wrist-based) and Polar H10 (chest-based).

6. Existing physiological data from both devices are used for evaluation.
7. The ability of each device to predict individual stress levels is assessed using a consistent analytical framework.
8. Device performance is validated through comparative evaluation, focusing on prediction accuracy and reliability rather than device advantages or disadvantages.

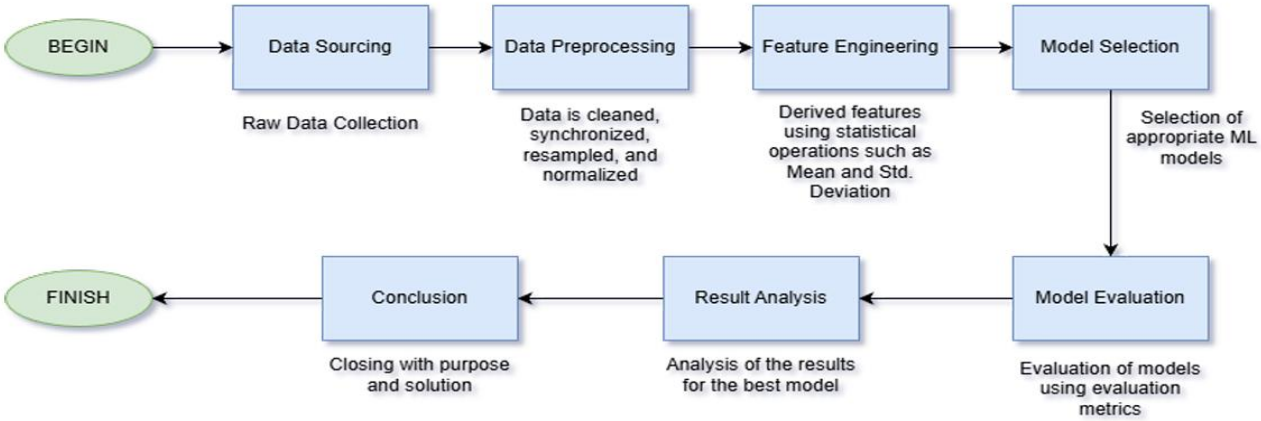


Fig. 1: Proposed machine-learning pipeline showing the end-to-end workflow of a sequential process.

4.1 Dataset Description and Preparation

4. 1.1 Dataset Description

For this study, the dataset is utilized from the GalaxyPPG dataset[19]. The dataset contains PPG signals recorded using three different wearable devices, namely Empatica E4, Polar H10, and Galaxy Watch. The dataset is deeply relevant to wearable-based stress monitoring studies. The dataset includes high-quality PPG signals captured from multiple participants, making it a valuable dataset for stress detection research.

This is a multimodal dataset recorded from 24 different participants using two wearable devices: a wristband and a chest strap. Data was recorded while the

participants performed three different activities-resting, cycling, and running-each at two different levels of intensity, hence covering all 4 movements considered light, moderate, and vigorous.

All these activities were performed continuously in 10-minute intervals. Various physiological and motion signals, such as heart rate (HR), blood volume pulse (BVP), accelerometer data (ACC), electrocardiogram (ECG), temperature (TEMP), and inter-beat interval (IBI), were captured by sensors on these devices

4.1.2 Data Preparation

Data pre-processing is a crucial task for ML learning. Data is cleaned, synchronized, and organized before modeling. Pre-processing prepares the raw sensor data for analysis by filtering the signals to reduce noise, resampling, and maintaining uniformity in the data. Data is normalized by standard scaling for better comparison of numbers

4.2 Feature Engineering

This study uses raw physiological data collected from wearable devices and transforms it into meaningful inputs to support effective model training. Raw signals are often long, noisy, and difficult for learning models to interpret directly. Such complexity can hide important physiological patterns related to stress. Feature engineering is therefore applied to reduce data complexity, remove noise, and extract informative characteristics that can be directly used by predictive models.

The wearable devices generate raw waveform signals such as ECG, heart rate, inter-beat interval (IBI), blood volume pulse (BVP), and accelerometer data. Since these waveforms are high-dimensional time-series data, additional features are derived to summarize their key properties. These derived features

allow the models to capture relevant physiological information without processing the entire raw signal.

• ecg mean, hr mean, ibi mean, bvp mean: the average values extracted using the features ECG, heart rate, IBI, and BVP.

• ecg std, hr std, ibi std, bvp std: represents standard deviation is extracted due to variability.

• ecg filtered, bvp filtered: the feature contains the cleaned ECG and BVP signals after removing noise.

• acc mag, acc mag smooth: the accelerometer data is interpreted and converted to ensure overall strength and smoothness of motion.

These features are primarily derived using statistical operations such as mean and standard deviation, along with signal filtering and magnitude computation. Filtering is applied to remove noise that could negatively impact model performance, while magnitude computation helps represent multi-axis motion data in a compact and interpretable form.

5.0 Experimental Setup and Implementation

The experiments are carried out using Python-based machine learning libraries. Physiological data collected from wearable devices are preprocessed to extract relevant features related to heart rate and heart rate variability. The dataset is divided into training and testing sets to ensure unbiased performance evaluation.

Model performance is assessed using standard classification metrics, including accuracy, Weighted F1-score. A comparative evaluation is performed across different machine learning models to analyze their effectiveness in accurately verifying stress levels.

6.0 Results and Analysis

This section presents the experimental evaluation of the proposed stress-level prediction framework using wearable physiological data. Model performance is evaluated using accuracy and weighted F1-score to account for class imbalance in stress labels. The results demonstrate the effectiveness of the proposed approach in distinguishing stress levels. The obtained metrics indicate that the model provides consistent and reliable stress-level classification performance, highlighting its suitability for practical stress monitoring applications.

In this study, the dataset initially consisted of raw signals with more noise, duplicates, and unscaled feature values,leading to undesired results. After a systematic data preprocessing, a series of machine learning models, including CatBoost Classifier, XGBoost Classifier, Deep MLP Classifier, LightGBM Classifier, Decision Tree Classifier, and Random Forest Classifier, are applied to develop an accurate predictive model. The results of the machine learning models are shown below

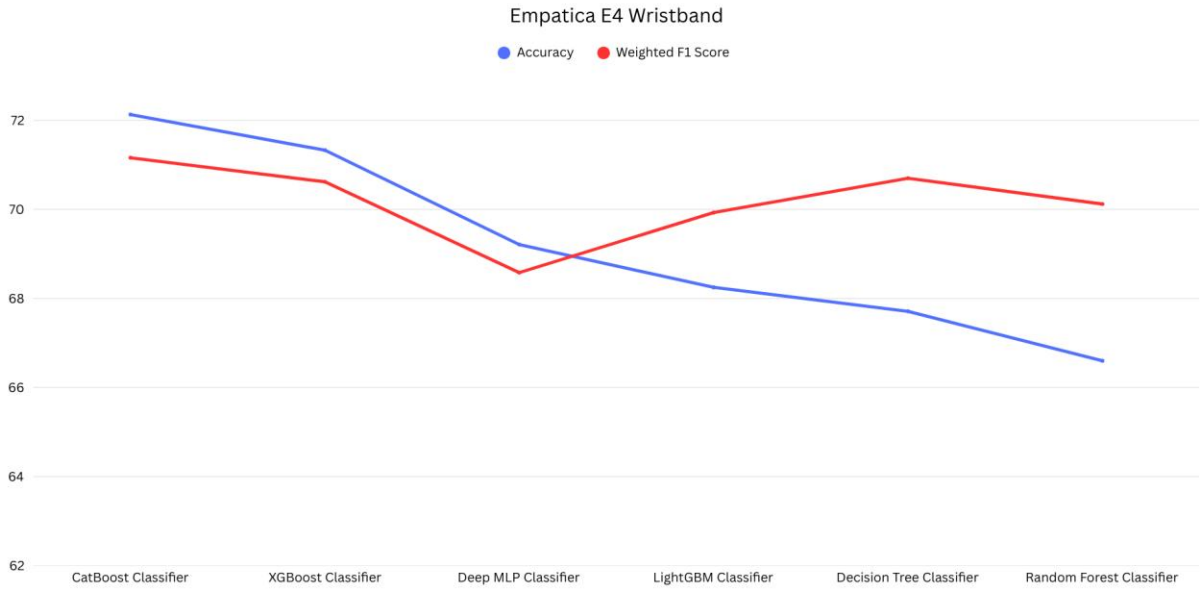
6.1 Performance of Machine Learning Models on Empatica E4 Dataset

ML Model	Accuracy (%)	Weighted F1 (%)
CatBoost Classifier	72.13	71.16
XGBoost Classifier	71.33	70.62
DeepMLP Classifier	69.21	68.58
LightGBM Classifier	68.25	68.93
DecisionTree Classifier	67.71	70.70
Random Forest Classifier	66.60	70.12

6.2 Performance of Machine Learning Models on Polar H10 Dataset

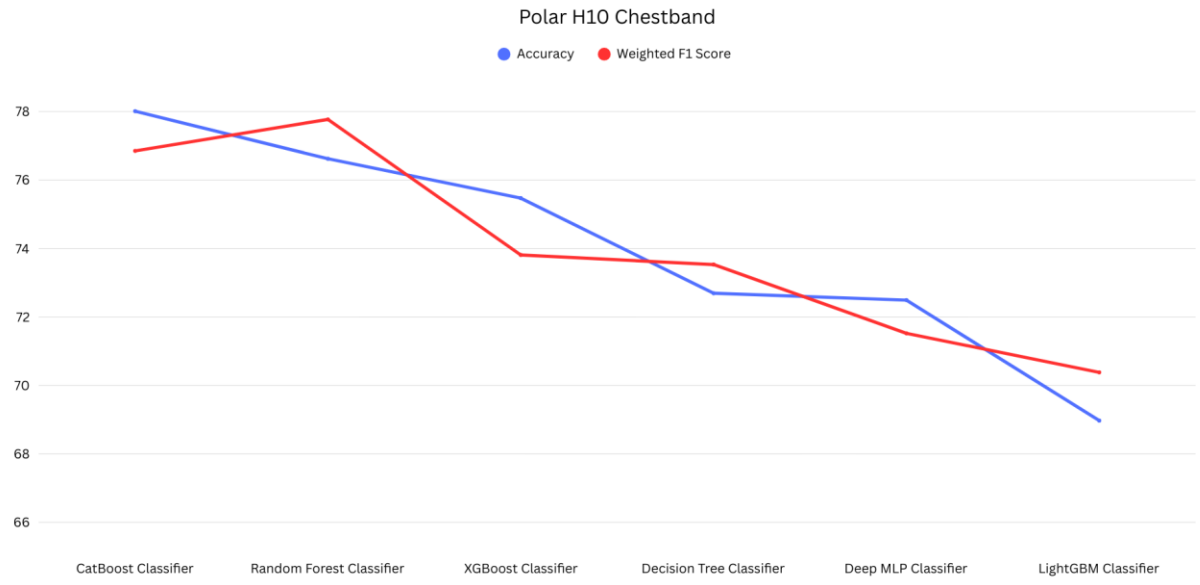
ML Model	Accuracy (%)	Weighted F1 (%)
CatBoost Classifier	78.01	76.85
XGBoost Classifier	75.47	73.81
DeepMLP Classifier	72.49	71.52
LightGBM Classifier	68.97	70.38
DecisionTree Classifier	72.69	73.53
Random Forest Classifier	76.62	77.77

6.3 Line chart of Accuracy and Weighted F1 score for Empatica E4 in comparison with machine learning models.



The figure compares the performance of individual machine learning classifiers using accuracy and weighted F1 score evaluation metrics on the Empatica E4 wristband dataset. It is observed that CatBoost and XGBoost yieldes higher results among these, yielding an accuracy above 70\%, while Deep MLP and LightGBM yieldes comparatively moderate performance. Decision tree and random forest showes lower accuracy and F1 scores, reflecting a lesser predictive capacity of these algorithms on the dataset.

6.4 Line chart of Accuracy and Weighted F1 score for Polar H10 in comparison with machine learning models.



The figure compares the performance of individual machine learning classifiers using accuracy and weighted F1 score evaluation metrics on the Polar H10 dataset. The figure shows that the highest accuracy and weighted F1 Score is for CatBoost, which is followed closely by Random Forest and XGBoost. In contrast to that, Decision Tree, Deep MLP, and LightGBM show lower performance values, reflecting dataset-specific variability in classifier effectiveness. These observations, taken together, convey the impression that gradient-boosting models-essentially

CatBoost and XGBoost-consistently outperform other stand-alone classifiers on both physiological sensing modalities.

7.0 Comparative Analysis with State-of-the-Art

Unlike Umair et al. (2021)[15], which focuses on comparing wearable sensor data quality and user perceptions without developing predictive models, our work directly builds and evaluates stress classification models using these physiological signals.

Their study reports correlation and agreement metrics between sensor measurements, whereas our research provides quantitative performance metrics (accuracy and weighted F1-score) for stress prediction, demonstrating practical classifier effectiveness.

8.0 Deployment and Practical Implementation

Input: Physiological signals from Empatica E4 (BVP, Temp, ACC) and Polar H10 (ECG, HR, IBI)

Processing: Pre-processing → Feature extraction → CatBoost stress-level classification

Output: Stress level prediction (Low / Medium / High)

Platform: Python-based system deployable on PC / laptop for research and monitoring

Use-Case: Supports real-time or offline stress analysis for health and wearable-device studies

9.0 Discussion

The experimental results demonstrate that machine learning models can effectively classify stress levels using physiological features derived from wearable devices. Among the evaluated models, ensemble-based approaches such as CatBoost and XGBoost achieved higher accuracy and weighted F1-scores, indicating their ability to handle complex feature interactions and class imbalance more effectively than simpler models. The use of weighted F1-score provides a balanced assessment of model performance across stress classes, ensuring that minority stress states are adequately represented. Overall, the results highlight the feasibility of using lightweight, feature-based machine learning models for reliable stress monitoring, making the proposed approach suitable for real-world wearable health applications.

10.0 Conclusion and Future Work

This study demonstrates the effectiveness of machine learning for stress level prediction using wearable device data. Among the evaluated models, CatBoost achieved the best performance, with the Polar H10 chest strap outperforming the Empatica E4 due to more stable sensing and the availability of ECG-based inter-beat interval measurements. Overall, the results highlight the importance of sensor quality and physiological signal selection in accurate wearable-based stress monitoring.

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